

CoBit / BitstreamDiffusion — Progress Report

Friday 4 – Wednesday 9 September 2026

Status against the original task list

Task	Status	Headline
1. Reproduce best baselines	DONE	replicated exactly; gap was churn alone
2. CFG + other guidance mechanisms, calibrate	DONE	CFG best, +0.081; study CLOSED
3. Nature of errors on TinyGSM	DONE	guidance harms code validity, not reasoning
4a. Temporal ordering — at inference	DONE	no benefit; random is harmful
4b. Temporal ordering — trained (L2R/R2L/random)	RUNNING	control arm defective; see §5
5. Logits: softmax vs score matching	DONE / RUNNING	softmax $\equiv$ BCE (proved); SM vs CE running

The most consequential finding of the week was not on the task list. Every training run in this environment diverged before $\sim 12k$ steps, which blocked all training experiments for four days. It is now solved: the cause is the **learning rate**, not the software environment. At $1r\ 3e-5$ a from-scratch run reached 20,000 steps cleanly where $3e-4$ diverges 8/8 times.

1 Task 1 — Reproduce best baselines DONE

Forensic replication audit against the collaborator's result: **verdict A, exact replication**. Our 0.1385 and their ~ 0.29 differ by **one factor: EDM stochastic churn**, worth +0.1622 and free in compute. Two churn code paths existed (`--gamma` vs `--churn_gamma`); the studies used different ones.

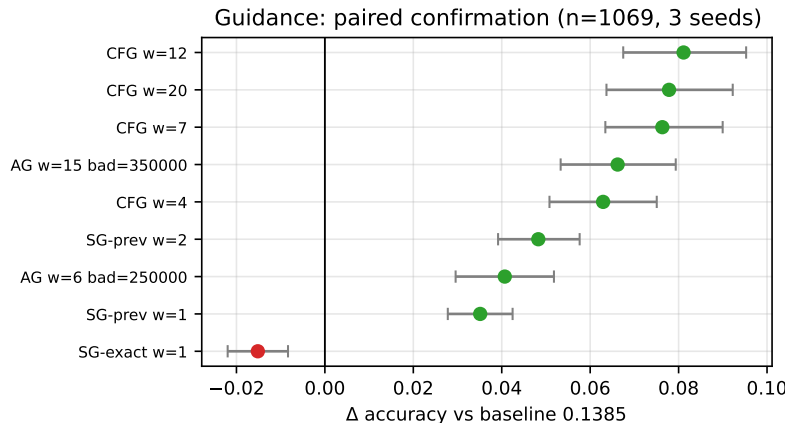
Baseline hierarchy on the 1319-problem GSM8K test set:

Setting	Accuracy	Note
Regime A (DDIM, $\gamma=0$, 500 steps)	0.1385	original arm
Regime B (FKC-EM, churn 0.41, 1024 steps)	~ 0.29	canonical, collaborator-matched
Production 500k, karras/DDIM, 256 steps	0.1640	anchor used throughout

Both regimes are retained and **every result is labelled with its regime**; they are never pooled.

2 Task 2 — Guidance mechanisms DONE (CLOSED)

Three mechanisms implemented and compared: classifier-free (CFG), AutoGuidance (AG), and Self-Guidance (SG-prev, SG-exact). Confirmation runs use 1069 problems $\times$ 3 seeds, paired bootstrap.



Config	Accuracy	Δ	95% CI	p	NFE
CFG $w=12$	0.2196	+0.0811	[+0.068, +0.095]	$< 10^{-5}$	1024
CFG $w=7$	0.2148	+0.0763	[+0.063, +0.090]	$< 10^{-5}$	1024
AG $w=15$ (bad@350k)	0.2047	+0.0662	[+0.053, +0.079]	$< 10^{-5}$	1024
SG-prev $w=2$	0.1868	+0.0483	[+0.039, +0.058]	$< 10^{-5}$	512
SG-exact $w=1$	0.1233	-0.0152	[-0.022, -0.008]	$< 10^{-5}$	1024

Findings. CFG is the strongest mechanism and saturates around $w \approx 7-12$. SG-prev gives about 60% of CFG’s gain at **half the NFE**, so it is the compute-efficient option. SG-exact actively hurts. **Mechanism:** guidance is not a correction — it fixes $\sim 10\%$ of problems and breaks $\sim 20-25\%$ regardless of regime, so its net value is arithmetic and only positive above a break-even baseline accuracy of $\approx 33\%$.

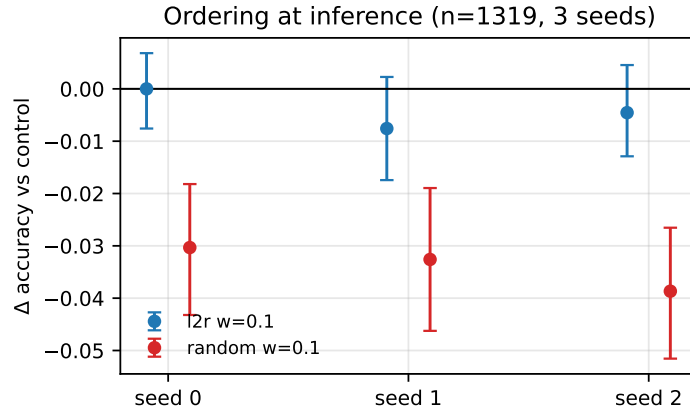
3 Task 3 — Nature of errors on TinyGSM **DONE**

Taxonomy: **T1** non-executing program, **T2** executes but wrong number, **T3** token-level degeneration.

The baseline’s dominant failure is T1 (98/250 = 39%) — the model emits programs that do not run. Under guidance, **T2 falls while T1 rises**: guidance damages *code validity*, not arithmetic reasoning. T3 appears only under strong guidance. Invalid-token rate is $3 \times 10^{-6} - 5 \times 10^{-5}$, i.e. bit $\rightarrow$ token decoding is not the problem.

4 Task 4a — Temporal ordering at inference **DONE**

Schedule: $t_j(t) = \text{clip}(t(1+w) - w u_j, 0, 1)$, with u_j a denoising *priority* over the generated **suffix only**; prompt keeps the global σ . Verified causally active: at $w=0.5$ the per-position σ spread is $192\times$, first suffix token cleanest; at $w=0$ it is bit-identical to the current model.



Arm	Accuracy	Δ	significant	Verdict
control ($w=0$)	0.1387	—	—	—
L2R $w=0.1$	0.1347	-0.0040	0/3 seeds	no effect
random $w=0.1$	0.1049	-0.0339	3/3 , $p < 10^{-4}$	harmful
L2R $w \geq 0.25$	0.0000	-0.176	3/3	collapse

Applying an ordering only at decoding does not help. The L2R/random asymmetry shows the intervention is real — direction matters — but nothing improves. Expected, since the model was *trained* with uniform σ : it tolerates a $\sim 2.9\times$ spread with no gain and breaks at $\sim 9\times$. 512 steps does not rescue it, so it is not a too-short-trajectory artefact.

5 Task 4b — Temporal ordering, **TRAINED RUNNING**

This is the substantive version of the experiment and it is **running, not finished**. Implementation is complete and tested (24 tests): σ is drawn per example, mapped to normalised time by the *same* map the sampler uses, offset per token, mapped back and expanded blockwise to bits. All three modes provably produce *different* σ fields; L2R and R2L are mirror images; random is redrawn per example *and* per step; $w=0$ leaves the gradient hash unchanged.

Four arms (control / L2R / R2L / random, $w=0.25$) fine-tune from the same production 500k checkpoint — from-scratch was not evaluable when these were launched. **Current blocker: the control arm is defective.** At the matched 5k checkpoint it scores 0.0000 while ordered arms score 0.07–0.12, which would read as “ordering massively helps” and is not credible. Its weights moved only 4.65% from init and its EMA is healthy, so the model is not destroyed; diagnosis is ongoing. **No ordering-training claim is made in this report.**

6 Task 5 — Logits: softmax vs score matching

Softmax is not a distinct objective, and this is proved rather than assumed DONE. The model already emits **one logit per bit** (`out_dim=1`, `vocab_size=2`). A 2-class softmax is shift-invariant, so it has a single degree of freedom $d = z_1 - z_0$ and

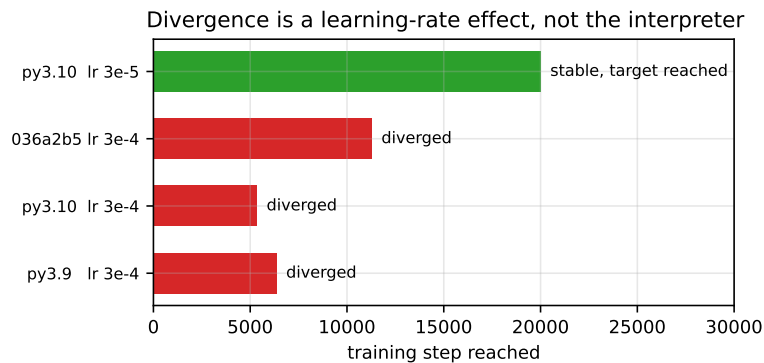
$$\text{softmax}(z_0, z_1)_1 = \sigma(d), \quad \text{CE}(z, y) = \text{BCEWithLogits}(d, y)$$

exactly; the 2-class head merely splits the same gradient across two logits ($\partial L / \partial z_1 = -\partial L / \partial z_0 = \partial L / \partial d$). Verified to 10^{-12} in 5 tests. **No GPU was spent on it.**

The meaningful comparison is **binary_sm vs binary_ce**, now **RUNNING** from scratch, 50k steps/arm, differing in exactly four keys (`train.loss_type` plus three output paths).

σ -weighting was derived, not inherited. Matching Bayes-risk scale gives a factor spanning only 0.250–0.361 across four decades of σ ($0.82\text{--}1.18\times$ anchored at σ_d); gradient-scale matching is self-defeating because it reintroduces the $D(1-D)$ factor under test. Conclusion: **no bespoke CE weighting is justified**; both arms use EDM.

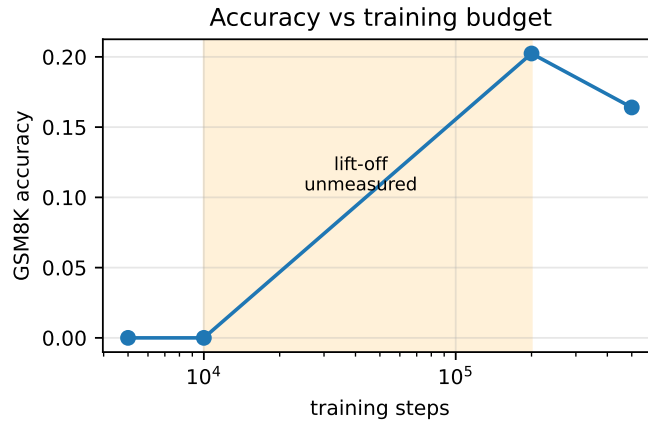
7 The infrastructural finding (not on the task list)



Eight from-scratch runs diverged before $\sim 12\text{k}$ steps at $1\text{r } 3\text{e-}4$, **including the production-era commit itself.** Ruled out along the way: corpus (σ_{data} fingerprint matches to ~ 8 s.f.), data config (identical on 22 keys), our post-June code (fp32 forward+loss+backward **bit-identical** to the production commit), and the Python interpreter — a 3.10 rebuild diverged at 5,346, falsifying a hypothesis I had put weight on.

At $1\text{r } 3\text{e-}5$ a from-scratch run reached 20,000 steps cleanly, as did all four ordering arms. The learning rate is the identified factor. *Caveat:* $3\text{e-}5$ establishes a **stable** regime, not a **tuned** one. It is $10\times$ below production’s value, so absolute accuracies in this regime are compared only against the matched control and never against production’s $3\text{e-}4$ trajectory.

8 Why accuracy could not be measured on short runs



GSM8K accuracy is **exactly 0.0000 at both 5k and 10k** training steps (measured, both objectives), and the first non-zero point we have is 0.2024 at 200k. Combined with divergence before 12k, the trainable range and the range where accuracy exists **did not overlap** — which is why the objective branch was blocked rather than merely slow.

9 Currently running

Job	ETA	Purpose
Ordering eval, 5k + 30k, karras	~1h	4 arms $\times$ 3 seeds $\times$ 1319
L2R retrain to 30k	~5h	its checkpoint was corrupted by a disk-full event
binary_sm vs binary_ce , from scratch, 50k	~7h	the definitive objective comparison

10 Not done

1. **Ordering-as-training verdict.** Blocked on the defective control arm.
2. **CE vs SM verdict.** Running; no result yet.
3. **Accuracy lift-off point** between 10k and 200k steps — unmeasured, and it sets the cost of every future training comparison.
4. **Why production tolerated $1r\ 3e-4$ for 500k steps.** Their working tree is permission-denied to us, so this stays open.
5. **Ordering at other strengths / from scratch**, and R2L at inference (deliberately not run: the first results did not justify it).

11 Honest notes

Three self-inflicted failures cost ~ 15 GPU-h and a night of wall-clock: an over-strict guidance guard failed 9/9 evaluation cells; the ordering control returned probabilities from σ_{N-1} instead of a final denoise, scoring 0.000 against a true 0.164; and a disk-full event killed four training arms mid-checkpoint. Each was caught before it entered a result. The second was caught *only* because a production anchor was included in the evaluation — without it, seven cells of confidently wrong nulls would have been reported.

A fourth was caught this morning: the entropic sampling schedule is fitted per-run, so each fine-tuned arm had refitted its own decoder schedule (max $|cdf - prod|$ up to 0.545). Evaluating that way compares arms differing in both training *and* decoding. All evaluations now use the analytic karras schedule, identical for every arm by construction.